

Podcasts and Review of Related Literature

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Today's students operate in a digital and global environment dominated by digital devices such as cell phones, personal digital assistants (PDAs), and mobile devices—and so does today's business (Coleman, 2005). Everything is technologically sophisticated and connected, including computers, cell phones, PDA, iPods, and other MP3 players. These digital devices are capable of receiving email, instant messages, discussion board posts, blogs, and Web conference messages. In recent years, these digital devices have undergone significant advances, enabling them to record and download podcasts as well as play audio and video. This is an area of particular concern for schools: they must be ready to respond to the cultural shifts of digital technology, and Holley (2006) has also echoed concerns about technology's impact on schools. Many other research studies on technology-based learning have also shown that students are likely to focus on the best resources to learn. Aytekin (2005) has stated that students learn best when they actively construct their own understanding in addition to reflective and more responsive forms of education.

The purpose of this examination is to determine whether podcasts can enhance student performance (or learning) and how they can be used to improve student outcomes. A research question: Does the use of podcasts improve students' math and reading achievement? In seeking answers to this question, it is important for us, as educators, to review prior studies and related literature on the relationship between podcasts and student performance. If we should, in fact, find a correlation between podcasts (or technology-based education) and student performance, we might then be interested in answering how podcasts can be used to enhance student learning.

One such study examines instructional technologies that support students' learning preferences. Students prefer the use of technology as a way to communicate with each other and with their instructor. Communication such as email, discussion boards, instant messaging, blogs, and conference systems has created more efficient and effective learning and teaching environments. Advances in communication technologies have made podcasting possible through personal digital devices such as iPods and other MP3 players (Ficek, 2006).

In recent years, we have seen that cell phones, PDAs, and the iPod (digital media player) are becoming much more advanced and more than just personal digital devices or media. These digital devices are now capable of receiving digital media such as video/audio podcasts distributed over the Internet for playback. Furthermore, these digital devices and the Internet are very much a part of people's lives.

How does podcasting change the way teachers teach and students learn? A podcast allows teachers to easily publish (or podcast) lectures or photos with a myriad of other course content. In addition, podcasts also enable students to listen to or view lectures and lessons on demand. Most students and teachers have access to cell phones and the Internet, and these digital devices are becoming a part of people's lives. For this reason, it is in the school's interest to determine how to integrate technology into the learning environment.

What is a podcast? A podcast is derived from Apple's portable music player, the iPod, which enables the delivery of educational content for listening or viewing on students' mobile devices, PDAs, and personal computers. Podcasting is a video/audio file feed in the Really Simple Syndication (RSS) format, a family of web feed formats used to publish frequently updated digital content, such as blogs or podcasts.

What is Podcasting? Podcasting refers to a set of technologies that facilitate the automatic distribution of audio and video programs over the Internet. It empowers independent creators to produce self-published, syndicated "radio shows" while also providing traditional radio and television programs with a new distribution channel. Any digital audio player or computer equipped with audio playback software can access podcasts. It's important to note that the term "podcast" primarily refers to audio content distribution and is distinct from a webcast, which typically refers to shows delivered via streaming media (<http://en.wikipedia.org/wiki/Podcast>).

Having spent many years in the technology industry, I have witnessed significant advancements in technology-based education over the past decade since its emergence. Today's technology-driven educational tools can encompass personal computers, PDAs, wireless networks, cell phones, iPods, and various software applications or digital devices. I firmly believe that technology has become an integral part of our lives, serving as a new medium for communication and connection. Additionally, I observe a growing trend in podcasts—an innovative form of video and audio content delivery that has evolved alongside digital technologies. Technology-based education and podcasts on mobile devices can effectively enhance teaching and learning experiences.

From the educational literature, Frey and Birnbaum (2002) indicate that computer-assisted instruction positively influences lectures, particularly in enhancing note-taking and exam preparation. Furthermore, research reveals that technology-based instructional methods are viewed favorably by students and lead to notable improvements in learning, as demonstrated by both self-reported feedback and objective test outcomes. (Smith & Woody, 2000).

The World Wide Web has fostered an information-rich environment for teaching and learning, enhancing social interaction and collaboration between students and instructors. It enables both teachers and students to access a vast array of knowledge-based resources and assessment tools that not only replicate but also expand upon the traditional classroom experience. This platform has demonstrated its effectiveness in storing, disseminating, and retrieving relevant course information, making it accessible to students at any time and from any location. (Aggarwal & Bento, 2000).

Given the diverse range of results, an increasing number of educators are incorporating technology-assisted instruction into their classrooms. Numerous research studies have suggested that technology-based learning can enhance student performance. However, the question of whether using podcasts improves students' math and reading achievement remains unanswered. This uncertainty stems, in part, from the fact that iPods and related MP3 players are still relatively new technologies; hence, additional research is needed to evaluate any potential long-term effects. Nevertheless, there is compelling evidence indicating a strong correlation between students' academic achievement and technology-based learning.

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